THE MODULI ALGEBRA MAP IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4645-4650 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4652-4713, the complete author proof (60 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_stacks-stacks-more-morphisms.json (pb=2649, stmt 4645-4650, proof 4652-4713).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-universally-injective' also exists in stacks-morphisms, stacks-spaces-morphisms — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4575–4583. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GRF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4713–4713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4713–4713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GQN: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4713–4713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4713–4713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Notation as in Lemma lemma-affine. Assume $\mathcal{X}$ is Noetherian and g is surjective and flat. Then $B \to C$ is universally injective.*

Proof. Consider the natural map $1 : \mathcal{O}_\mathcal{X} \to g_{QCoh,*}\mathcal{O}_U$ in $QCoh(\mathcal{O}_\mathcal{X})$. Pulling back to U and using adjunction we find that the composition

$$\mathcal{O}_U = g^*\mathcal{O}_\mathcal{X} \xrightarrow{g^*1} g^*g_{QCoh,*}\mathcal{O}_U \to \mathcal{O}_U$$

is the identity in $QCoh(\mathcal{O}_U)$. Write $g_{QCoh,*}\mathcal{O}_U = \operatorname{colim} \mathcal{F}_i$ as a filtered colimit of coherent $\mathcal{O}_\mathcal{X}$-modules, see Cohomology of Stacks, Lemma 0GRF. For i large enough the map $1 : \mathcal{O}_\mathcal{X} \to g_{QCoh,*}\mathcal{O}_U$ factors through $\mathcal{F}_i$, see Cohomology of Stacks, Lemma 0GQZ. Say $s : \mathcal{O}_\mathcal{X} \to \mathcal{F}_i$ is the factorization. Then

$$\mathcal{O}_U \xrightarrow{g^*s} g^*\mathcal{F}_i \to g^*g_{QCoh,*}\mathcal{O}_U \to \mathcal{O}_U$$

is the identity. In other words, we see that s becomes the inclusion of a direct summand upon pullback to U. Set $\mathcal{F}_i^\vee = hom(\mathcal{F}_i, \mathcal{O}_\mathcal{X})$ with notation as in Cohomology of Stacks, Lemma 0GQN. In particular there is an evaluation map $ev : \mathcal{F}_i \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{F}_i^\vee \to \mathcal{O}_\mathcal{X}$. Evaluation at s defines a map $s^\vee : \mathcal{F}_i^\vee \to \mathcal{O}_\mathcal{X}$. Dual to the statement about s we see that $g^*(s^\vee)$ is surjective, see see Cohomology of Stacks, Section 0GQS for compatibility of hom and $\otimes$ with restriction to U. Since g is surjective and flat, we conclude that $s^\vee$ is surjective (see locus citatus). Since F is right exact, we conclude that $F(\mathcal{F}_i^\vee) \to F(\mathcal{O}_\mathcal{X}) = B$ is surjective. Choose $\lambda \in F(\mathcal{F}_i^\vee)$ mapping to $1 \in B$. Denote $e = F(s)(1) \in F(\mathcal{F}_i)$ the image of 1 by the map $F(s) : B = F(\mathcal{O}_\mathcal{X}) \to F(\mathcal{F}_i)$. Then the map

$$F(ev) : F(\mathcal{F}_i) \otimes_B F(\mathcal{F}_i^\vee) = F(\mathcal{F}_i \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{F}_i^\vee) \longrightarrow F(\mathcal{O}_\mathcal{X}) = B$$

sends $e \otimes \lambda$ to 1 by construction. Hence the map $B \to F(\mathcal{F}_i)$, $b \mapsto be$ is universally injective because we have the one-sided inverse $F(\mathcal{F}_i) \to B$, $\xi \mapsto F(ev)(\xi \otimes \lambda)$. Since this is true for all i large enough we conclude. □